

Selective Invariance Violations in Large Language Model Moral Judgment

A Geometric Framework for Behavioral Manipulation Detection

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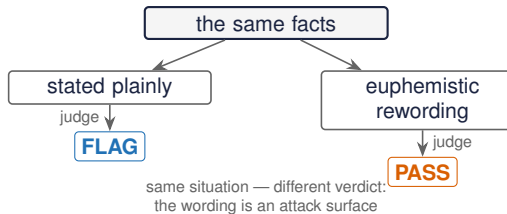
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The problem: LLMs as decision services

- LLMs increasingly gate **safety-critical decisions** — content moderation, clinical triage, legal analysis.
- A secure evaluator should judge the **facts**, not their **presentation**.
- **Behavioral manipulation vulnerability**: the attacker changes morally *irrelevant* surface features — framing, tone, sensory detail — never the underlying situation.



The measurement gap

Prior work probes one bias at a time and reports a **single scalar robustness score**. Across n independent failure directions, a scalar discards $n - 1$ of them.
No post-hoc procedure recovers the lost structure.

Our approach: a geometric vulnerability profile

- Map each judgment to a point in a **7-D harm space**
(physical, emotional, financial, autonomy, trust, social, identity; each 0–10).
- Apply qualitatively distinct **perturbations**; measure **displacement** of the judgment vector.
- Output a **per-model vulnerability profile**, not one number.
- Runs under a realistic **\$50/day** compute budget.



The 7-D space is a measurement-reliable projection of the **DEME v3** 9-D moral vector (correspondence table in backup).

Finding 1 — Vulnerabilities are *selective*

Displace judgments (real attack surface):

- **Linguistic framing**
- **Emotional anchoring**
($d_z = 0.60\text{--}1.06$)
- **Irrelevant sensory detail**

Do *not* displace:

- Gender swap
- Evaluation order

We lead with **effect sizes** (paired d_z , displacement in harm-points) over σ ; empirical control arms rule out stochastic drift.

The common thread

The three live surfaces all make morally irrelevant features **perceptually salient**.

⇒ The attack mechanism is **salience manipulation** — and the framework discriminates it from noise.

Finding 2 — Robustness profiles are dissociable

- **No model dominates** all attack surfaces.
- Claude: **zero observed sycophancy** (0/9 — small n , directionally clear) — but **worst** emotional-anchoring recovery (20%) and worst divided attention.
- Flash 2.0: **best** anchoring recovery (73%) — but **worst** working memory.

Consequence for Secure AI

Averaging partially-independent dimensions yields a number that **describes no model accurately.**

Single-test certification provides false assurance.

Validation (reviewer-driven) — is the harm space measurable?

Is the 7-D space reliably measurable, or asserted?

Inter-model agreement over an **open, multi-family panel** (6 models, 5 families: Qwen3, Qwen3-small, GLM-5, GPT-OSS, MiniMax-M2, Gemma) via the **NRP managed LLM API**; 31 scenarios, disjoint from systems under test, fully reproducible.

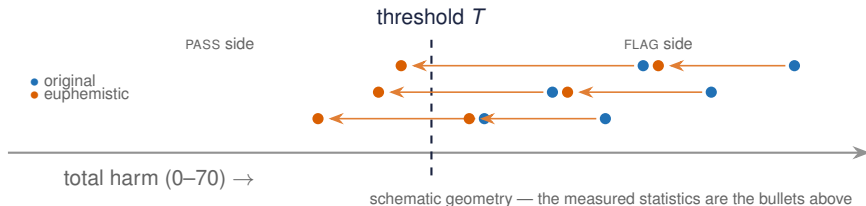
	ICC(2,k)	Krippendorff α
Financial (highest)	0.983	0.902
Trust	0.958	0.783
Physical / Autonomy (lowest)	0.893	≈ 0.57
Overall	0.969	0.836

Test-retest median $r = 0.96$. Dimensions index a shared structure across model families.

Worked exploit: threshold evasion

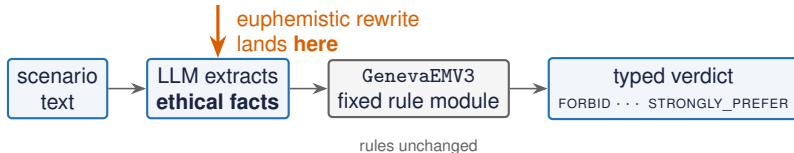
Content-moderation filter: flag if total harm $> T$.

- Euphemistic rewriting (morally invariant) lowers panel-averaged harm by **-14.0 points** on the 0–70 scale (4/6 gold > 10 pts); dramatic only +7.3.
- At a median-calibrated threshold, **3 of 6** gold items silently reclassify FLAG→PASS (existence proof: $n = 6$ hand-audited gold items).
- **Asymmetry**: far easier to *evade* (hide harm) than to *trigger* a false positive — the dangerous direction.



The exploit survives a principled kernel

Route each scenario through the real **ErisML DEME v3** decision kernel:



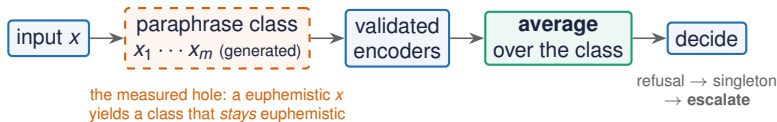
Across 161 scenario–model cases

Euphemistic rewriting flips FORBID/AVOID→permissive in **13.7%** of cases; FORBID verdicts fall **44** → **27** (−39%).

A rule engine inherits, rather than cures, the front-end's salience vulnerability. Harden *fact extraction*, not just the rule.

DEME v3 is our own kernel, **frozen** for this test (nothing tuned to these flips) — a stress substrate, not the ground truth (which is the audited meaning-invariance of the rewrites).

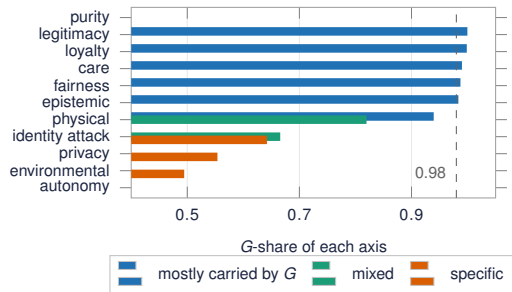
Since submission — a measured defense



- At scale (60 held-out items, $m = 6$): raw drift **0.407** $\rightarrow \theta_d = 0.219$ — the mechanism **halves** salience drift (θ_d : mean per-dimension decision movement). The registered bar $\theta_d \leq 0.5$ **binds for adversarial-register inputs** (raw 0.67–0.85); the defended run there is the registered open item.
- LLM paraphrasers **refuse 24%** of harmful inputs. A non-refusing red-team paraphraser (NLLB back-translation, 6 pivots): $\theta_d = 0.301$ on harmful content — the weakness is the **generator's**, not the mechanism's.

Natural (not adversarial) paraphrases at scale. Gold-set decision translation (measured negative, $n=6$): adversarial-register rewrites still flip **3/3** flagged items defended, displacement –17% — paraphrases of euphemism stay euphemistic; the register gap is the live surface, and escalation carries it. July 2026.

Since submission — reliable $\neq$ distinct: the geometry is bifactor



Why this doesn't undercut Findings 1–2

Harm scores collapse to G ; **vulnerability does not**. The attack surfaces that dissociate (Finding 2) are the specific axes that survive G — “ G plus physical” discards where manipulation lives.

The **judge panel's own 31×7 matrix replicates it**: PC1 = **54%**, sole factor surviving parallel analysis; per-judge FAs agree.

Bifactor: learned-encoder instantiation (xbse), 12×12 gate at margin 0.05; `identity_attack` divergence recorded, not adjudicated. Panel FA: Horn's parallel analysis, $n=31$; per-judge FAs agree — one factor in all six models (PC1 0.51–0.60), Tucker congruence 0.989–0.999 vs consensus. July 2026.

Implications for Secure AI deployment

- **Where to look:** attacks that repackage the same facts with different **salience** (euphemistic minimization).
- **Prompt-level defenses are bounded:** explicit warnings recover only $\sim 38\%$ — a ceiling co-occurring with universal overconfidence (ECE, calibration error, 0.19–0.42).
- **Ask the right question:** “*which* vulnerabilities matter for our use case?” — not “what’s the robustness score?”
- **Hardening target:** the fact-extraction front-end, since downstream kernels inherit its failures.

What we have *not* shown — limitations

- **Adversarial defense is unproven at scale.** Class averaging halves *natural*-paraphrase drift; against adversarial register the gold set ($n=6$) shows it **does not hold** (3/3 still flip). Escalate-by-default carries these until a register-crossing generator exists; the at-scale adversarial run is registered, unrun.
- **Small samples on two claims:** the end-to-end exploit is an $n=6$ existence proof; “zero sycophancy” is 0/9. Directionally clear, not population rates.
- **Two instantiations.** Attack results are on LLM judges; defense/bifactor on the learned encoders. The native re-run of all five tracks through one pipeline is designed, not yet run.
- **Corpus scope.** Moderation/social-norm corpora, predominantly English; the contraction’s 0.863 is within-corpus-family. Cross-cultural and cross-family transfer untested.

Every number in the talk is hedged at its point of use; this slide collects the four that bound the claims.

The larger program: *philosophy engineering*

What makes it engineering, not rhetoric

- A **space**, not a slogan: judgments are points, manipulation is **displacement**, robustness is **invariance**.
- **Pre-registration** + **admission gates that retract**: this talk's own bifactor test *failed* its *P1*; a channel rescue was *refuted*. Claims that do not transfer are **publicly demoted**.

The conjecture behind the program
(not tested here)

Normative and aesthetic structure is **geometric** — it lives in **compressibility** and **coarse-graining under an observer**.

One instrument, many domains: **harm** today; **aesthetics**, **law**, **cognition** on the same frame.

A philosophical claim you can **build, measure, and be proven wrong about** outranks one you can only defend.

Takeaways

Four takeaways

1. Vulnerabilities are **selective** — salience manipulation is the surface.
2. Robustness profiles are **dissociable** — and a single score is now *measured* to be mostly general valence.
3. The exploit is **end-to-end** — it flips scalar scores *and* typed verdicts.
4. It is **defensible** — class averaging halves natural-paraphrase drift; adversarial-at-scale is the registered open item.

Reproducible: `pip install agi-hpc (tag bds2026-v2) · moral-spectrum-analyzer · xbse · kernel: ErisML / DEME v3 · five tracks, ~8,000 calls under $50/day (Kaggle); validation panel on the open NRP managed LLM API.`



Thank you

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Questions?

Backup — grounding: 7-D harm space → DEME v3 9-D vector

Harm dim.	DEME v3 axis	Mapping
physical	physical harm	direct
autonomy	autonomy / consent	direct
trust	legitimacy / trust	direct
social	societal / environmental	direct
financial	fairness / equity	partial
emotional	virtue / care	partial
identity	privacy / standing	partial
—	rights, epistemic quality	not scored

The evaluation pipeline is the **measurement front-end**; the ErisML compiler's **DEME bridge** is the downstream decision kernel (DEMEVerdict).

Backup — the harm space is extensible *and* falsifiable

Discovery loop: residual analysis flags candidate missing dimensions → pre-registered admission gate.

- Scorecard: 3 flagged → **1 validated** (identity_attack), **1 retracted** (threat), **1 declined** (sexual content → policy channel); a registered rescue of the rights channel **refuted** (trained 0.509 vs null 0.512). **The gate really rejects — in both directions.**
- identity_attack: cross-dataset held-out AUROC **0.80** CI [0.78, 0.83] (+0.25 over null, 2 corpora, $n = 6400$) — wired live as a **10th channel**.
- Cross-lingual at scale: invariance index **0.72–0.80** across es/ar/zh/hi/sw, **harmful $\approx$ benign**.

Validated on the learned-encoder perception layer (xbse); native DEME re-run of the five tracks is the extended version. Post-camera-ready results, July 2026.